Meta-Regression

This vignette introduces the regression functionality of the `dtametaTMB` package for meta-analysis of diagnostic test accuracy (DTA) studies.

Validation

The subgroup HSROC implementation reproduces the Cochrane Handbook RF example closely, including the baseline HSROC parameters (accuracy, threshold, shape), between-study variance estimates, and subgroup effects on accuracy and threshold. Small numerical differences may arise from optimizer settings, numerical tolerances, and implementation details, but all benchmark estimates and standard errors were recovered to close agreement.

Dummy-coding vs. cell-means parameterization

For subgroup analyses, we distinguish between two equivalent parameterizations of the same model. In the reference-group parameterization, one subgroup serves as the baseline and the remaining subgroup effects are expressed as deviations from this reference. In the group-specific (cell-means) parameterization, each subgroup has its own parameter directly. The former is convenient for testing subgroup differences, whereas the latter is convenient for reporting subgroup-specific estimates and plotting subgroup-specific HSROC curves. The Reitsma subgroup model is fitted twice with both parameterizations. The subgroup HSROC model is estimated using a reference-group parameterization, and subgroup-specific parameters are recovered by evaluating the fitted linear predictors at subgroup-specific design points via Z_pred.

How do we include covariates in the Reitsma model?

For sensitivity in study i, we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

ow we introduce a p-dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\begin{pmatrix} z_i^\top \mu_{Ai} \\ z_i^\top \mu_{Bi} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} z_i^\top \mu_A \\ z_i^\top \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

Let's assume that z_i includes a single binary covariate representing two subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \mu_{Ai} = \begin{cases} \mu_{Ai} & \text{for subgroup 1,} \\ \mu_{Ai} + \nu_{A2} & \text{for subgroup 2,} \end{cases} \qquad z_i^\top \mu_{Bi} = \begin{cases} \mu_{Bi} & \text{for subgroup 1,} \\ \mu_{Bi} + \nu_{B2} & \text{for subgroup 2,} \end{cases}$$

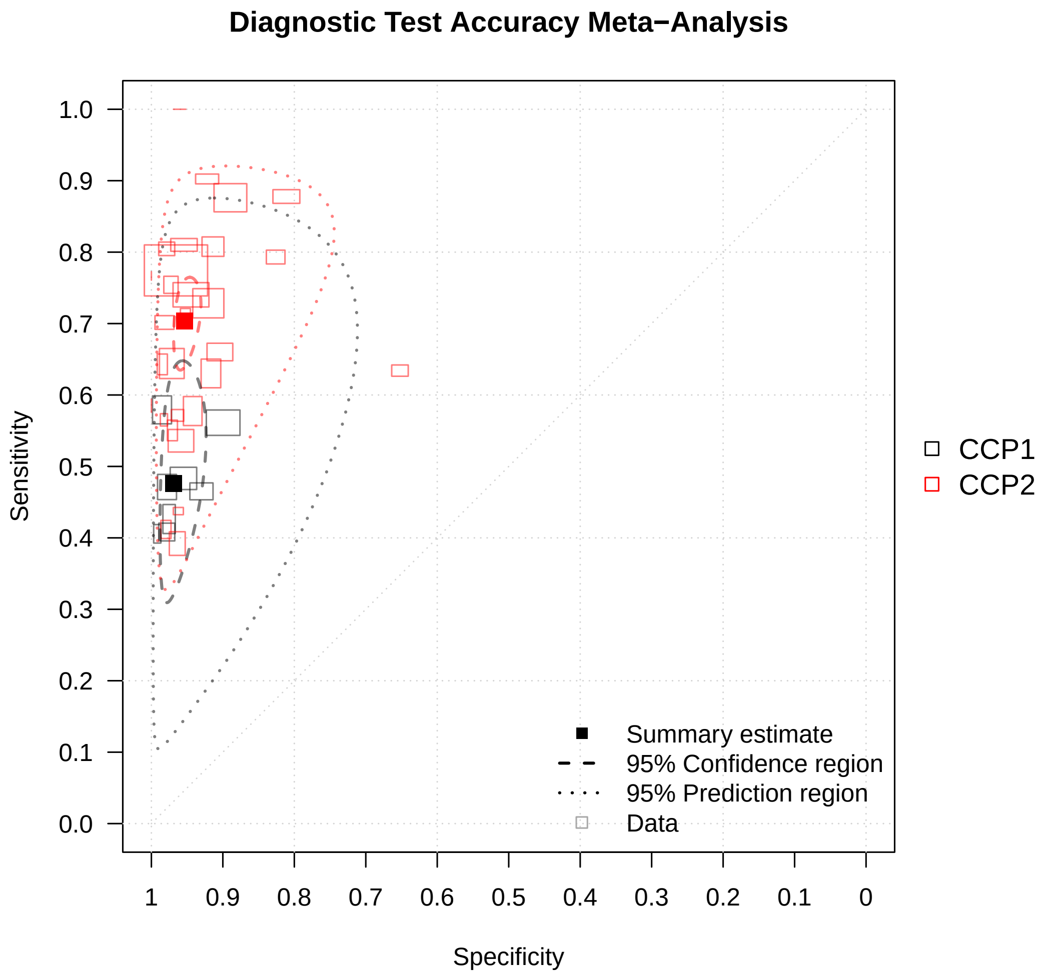
$$z_i^\top \mu_A = \begin{cases} \mu_A & \text{for subgroup 1,} \\ \mu_A + \nu_{A2} & \text{for subgroup 2,} \end{cases} \qquad z_i^\top \mu_B = \begin{cases} \mu_B & \text{for subgroup 1,} \\ \mu_B + \nu_{B2} & \text{for subgroup 2.} \end{cases}$$

```
data("anticcp")
reitsmasub <- fitReitsmaSubgroup(data=anticcp,
                                 TP=TP,
                                 FP=FP,
                                 FN=FN,
                                 TN=TN,
                                 study=study,
                                 subgroup=generation)
reitsmasub
#>
#> Reitsma Subgroup Model
#> ----------------------
#>
#> Number of studies   : 37
#> Number of subgroups : 2
#> Model fit           : Converged
#> -2 log likelihood   : 533.37
#>
#>
#> Use summary() for parameter estimates.
summary(reitsmasub)
#> $estimates
#>                  Estimate  Std_Error
#> mu_A.CCP1     -0.09654001 0.22032136
```

```
#> mu_B.CCP1      3.44670662 0.29824593
#> mu_A.CCP2      0.86603422 0.12087713
#> mu_B.CCP2      3.01649071 0.16223893
#> sigma2_A.sens  0.35983136 0.10217750
#> sigma2_B.spec  0.53992184 0.18016162
#> sigma_AB      -0.19685042 0.09835584
#> nu_A.CCP2      0.96255893 0.25134200
#> nu_B.CCP2     -0.43020026 0.33771277
#>
#> $sensspec
#>           type      Orig conflevel  CI_Lower  CI_Upper
#> mu_A.CCP1 sens 0.4758837      0.95 0.3708990 0.5830440
#> mu_B.CCP1 spec 0.9691328      0.95 0.9459436 0.9825577
#> mu_A.CCP2 sens 0.7039198      0.95 0.6522898 0.7508123
#> mu_B.CCP2 spec 0.9533136      0.95 0.9369386 0.9655927
#>
#> $RutterGatsonis_recovered
#>        Lambda    Theta      beta sigma2_alpha sigma2_theta
#> CCP1 4.121053 1.278040 0.2028944    0.4878451    0.3188117
#> CCP2 3.727490 1.950972 0.2028944    0.4878451    0.3188117
```

How do I get a summary plot of the Reitsma model?

```
plot(reitsmasub,
     scale=0.01,
     nudge_legend=-0.2,
     size="se",
     col=c("black","red"))
```

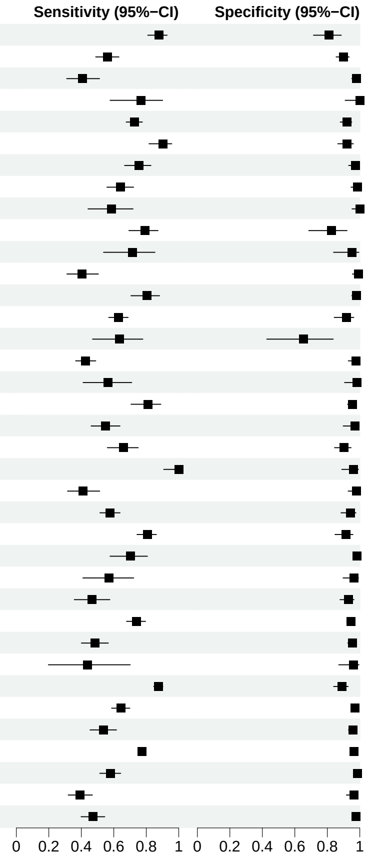


How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(reitsmasub,subgroup_label="Generation")
```

Study	Generation	TP	FP	FN	TN	Sensitivity (95%−CI)	Specificity (95%−CI)
Aotsuka 2005	CCP2	115	17	16	73	0.88 [0.81, 0.93]	0.81 [0.71, 0.89]
Bas 2003	CCP1	110	24	86	215	0.56 [0.49, 0.63]	0.90 [0.85, 0.93]
Bizzaro 2001	CCP1	40	5	58	227	0.41 [0.31, 0.51]	0.98 [0.95, 0.99]
Bombardieri 2004	CCP2	23	0	7	39	0.77 [0.58, 0.90]	1.00 [0.91, 1.00]
Choi 2005	CCP2	236	20	88	231	0.73 [0.68, 0.78]	0.92 [0.88, 0.95]
Correa 2004	CCP2	74	11	8	130	0.90 [0.82, 0.96]	0.92 [0.86, 0.96]
De Rycke 2004	CCP2	89	4	29	142	0.75 [0.67, 0.83]	0.97 [0.93, 0.99]
Dubucquoi 2004	CCP2	90	2	50	129	0.64 [0.56, 0.72]	0.98 [0.95, 1.00]
Fernandez−Suarez 2005	CCP2	31	0	22	75	0.58 [0.44, 0.72]	1.00 [0.95, 1.00]
Garcia−Berrocal 2005	CCP2	69	8	18	38	0.79 [0.69, 0.87]	0.83 [0.69, 0.92]
Girelli 2004	CCP2	25	2	10	40	0.71 [0.54, 0.85]	0.95 [0.84, 0.99]
Goldbach−Mansky 2000	CCP1	43	1	63	120	0.41 [0.31, 0.51]	0.99 [0.95, 1.00]
Greiner 2005	CCP2	70	5	17	228	0.80 [0.71, 0.88]	0.98 [0.95, 0.99]
Grootenboer−Mignot 2004	CCP2	167	8	98	88	0.63 [0.57, 0.69]	0.92 [0.84, 0.96]
Hitchon 2004	CCP2	26	8	15	15	0.63 [0.47, 0.78]	0.65 [0.43, 0.84]
Jansen 2003	CCP1	110	3	148	118	0.43 [0.37, 0.49]	0.98 [0.93, 0.99]
Kamali 2005	CCP2	26	1	20	56	0.57 [0.41, 0.71]	0.98 [0.91, 1.00]
Kumagai 2004	CCP2	64	14	15	293	0.81 [0.71, 0.89]	0.95 [0.92, 0.97]
Kwok 2005	CCP2	71	2	58	66	0.55 [0.46, 0.64]	0.97 [0.90, 1.00]
Lee 2003	CCP2	68	14	35	132	0.66 [0.56, 0.75]	0.90 [0.84, 0.95]
Lopez−Hoyos 2004	CCP2	38	3	0	73	1.00 [0.91, 1.00]	0.96 [0.89, 0.99]
Nell 2005	CCP2	42	2	60	96	0.41 [0.32, 0.51]	0.98 [0.93, 1.00]
Nielen 2005	CCP2	149	7	109	114	0.58 [0.51, 0.64]	0.94 [0.88, 0.98]
Quinn 2006	CCP2	147	10	35	106	0.81 [0.74, 0.86]	0.91 [0.85, 0.96]
Rantapaa−Dahlqvist 2003	CCP2	47	7	20	375	0.70 [0.58, 0.81]	0.98 [0.96, 0.99]
Raza 2005	CCP2	24	3	18	79	0.57 [0.41, 0.72]	0.96 [0.90, 0.99]
Saraux 2003	CCP1	40	11	46	146	0.47 [0.36, 0.58]	0.93 [0.88, 0.96]
Sauerland 2005	CCP2	171	26	60	443	0.74 [0.68, 0.80]	0.94 [0.92, 0.96]
Schellekens 2000	CCP1	72	14	77	298	0.48 [0.40, 0.57]	0.96 [0.93, 0.98]
Soderlin 2004	CCP2	7	2	9	51	0.44 [0.20, 0.70]	0.96 [0.87, 1.00]
Suzuki 2003	CCP2	481	23	68	185	0.88 [0.85, 0.90]	0.89 [0.84, 0.93]
Vallbracht 2004	CCP2	190	12	105	408	0.64 [0.59, 0.70]	0.97 [0.95, 0.99]
van Gaalen 2005	CCP2	82	13	71	301	0.54 [0.45, 0.62]	0.96 [0.93, 0.98]
van Venrooij 2004	CCP2	865	79	252	2218	0.77 [0.75, 0.80]	0.97 [0.96, 0.97]
Vincent 2002	CCP1	139	7	101	464	0.58 [0.51, 0.64]	0.99 [0.97, 0.99]
Vittecoq 2004	CCP2	69	5	107	133	0.39 [0.32, 0.47]	0.96 [0.92, 0.99]
Zeng 2003	CCP1	90	7	101	313	0.47 [0.40, 0.54]	0.98 [0.96, 0.99]



How do we include covariates in the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

Now we introduce a p-dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\mathrm{logit}(\pi_{ij}) = \left(z_i^\top \theta_i + z_i^\top \alpha_i x_{ij}\right) \exp(-z_i^\top \beta x_{ij}),$$

$$z_i^\top \alpha_i \sim \mathcal{N}(z_i^\top \Lambda, \sigma_\alpha^2), \qquad z_i^\top \theta_i \sim \mathcal{N}(z_i^\top \Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

Let's assume that z_i includes a single categorical covariate representing three subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \theta_i = \begin{cases} \theta_i & \text{for subgroup 1,} \\ \theta_i + \gamma_2 & \text{for subgroup 2,} \\ \theta_i + \gamma_3 & \text{for subgroup 3,} \end{cases} \qquad z_i^\top \alpha_i = \begin{cases} \alpha_i & \text{for subgroup 1,} \\ \alpha_i + \xi_2 & \text{for subgroup 2,} \\ \alpha_i + \xi_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \beta = \begin{cases} \beta & \text{for subgroup 1,} \\ \beta + \delta_2 & \text{for subgroup 2,} \\ \beta + \delta_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \Lambda = \begin{cases} \Lambda & \text{for subgroup 1,} \\ \Lambda + \xi_2 & \text{for subgroup 2,} \\ \Lambda + \xi_3 & \text{for subgroup 3,} \end{cases} \qquad z_i^\top \Theta_i = \begin{cases} \Theta & \text{for subgroup 1,} \\ \Theta + \gamma_2 & \text{for subgroup 2,} \\ \Theta + \gamma_3 & \text{for subgroup 3.} \end{cases}$$

Notes:

- By default, `fitRutterGatsonisSubgroup()` assumes a common shape across all subgroups, which implies $\delta_2 = \delta_3$ for the case above.
- Moreover, for prediction `fitRutterGatsonisSubgroup()` uses the prediction matrix

  $$Z_{\text{pred}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

  for the case above and therefore recovers the threshold, accuracy, and shape parameters for each subgroup as if it were the reference group.

How do I perform a Rutter and Gatsonis meta-regression with a single categorical study-level covariate?

```
data("RF")
RF2         <- RF[RF$method %in% c("LA","ELISA","Nephelometry"),]
RF2$method <- factor(RF2$method,levels=c("LA","ELISA","Nephelometry"))
ruttergatsonissub <- fitRutterGatsonisSubgroup(data=RF2,
```

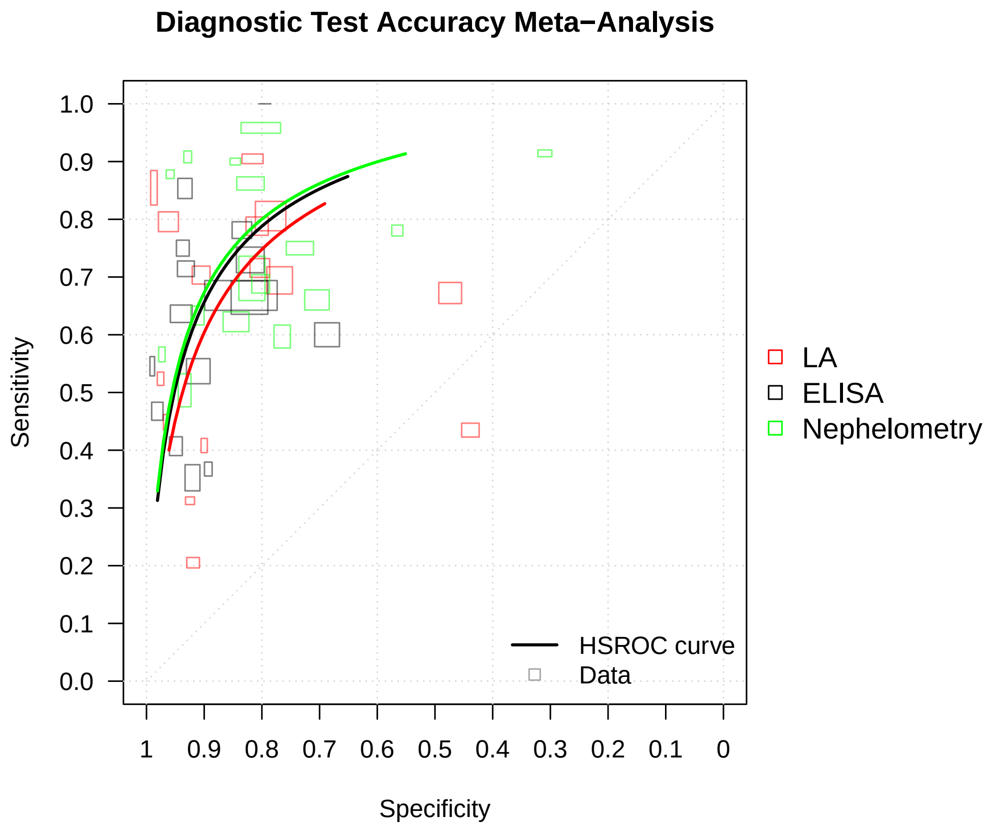
```
                                              TP=TP,
                                              FP=FP,
                                              FN=FN,
                                              TN=TN,
                                              study=study,
                                              subgroup=method)
```

```
ruttergatsonissub
#>
#> Rutter & Gatsonis Subgroup Model
#> --------------------------------
#>
#> Number of studies   : 47
#> Number of subgroups : 3
#> Model fit           : Converged
#> -2 log likelihood   : 753.722
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissub)
#> $estimates
#>                        Estimate Std. Error
#> Lambda_LA            2.45630704 0.32357707
#> xi_ELISA             0.24853054 0.43954126
#> xi_Nephelometry      0.33141751 0.44260069
#> Theta_LA            -0.55005213 0.21334057
#> gamma_ELISA         -0.19625909 0.26096239
#> gamma_Nephelometry   0.49586320 0.26223094
#> beta_LA              0.19768314 0.16983954
#> delta_ELISA          0.00000000 0.00000000
#> delta_Nephelometry   0.00000000 0.00000000
#> sigma2_alpha         1.27791582 0.30852514
#> sigma2_theta         0.47684216 0.11344574
#> Lambda_LA            2.45630704 0.32357707
#> Lambda_ELISA         2.70483758 0.32695434
#> Lambda_Nephelometry  2.78772454 0.30577914
#> Theta_LA            -0.55005213 0.21334057
#> Theta_ELISA         -0.74631122 0.20991330
#> Theta_Nephelometry  -0.05418893 0.21213149
#> beta_LA              0.19768314 0.16983954
#> beta_ELISA           0.19768314 0.16983954
#> beta_Nephelometry    0.19768314 0.16983954
```

```
#> logitsens           0.82616754 0.28629967
#> logitsens           1.05130793 0.28606921
#> logitsens           1.12639409 0.27689507
#> sens                0.69554397 0.06062755
#> sens                0.74102598 0.05489854
#> sens                0.75517283 0.05119425
#>
#> $sensspec
#>      subgroup      spec conflevel logitsens Std_Error  CI_Lower CI_Upper
#> 1           LA 0.8461538      0.95 0.8261675 0.2862997 0.2650305 1.387305
#> 2        ELISA 0.8461538      0.95 1.0513079 0.2860692 0.4906226 1.611993
#> 3 Nephelometry 0.8461538      0.95 1.1263941 0.2768951 0.5836897 1.669098
#>        Sens SensCI_Lower SensCI_Upper
#> 1 0.6955440    0.5658725    0.8001616
#> 2 0.7410260    0.6202531    0.8336879
#> 3 0.7551728    0.6419160    0.8414556
#>
#> $Reitsma_recovered
#>              mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec   sigma_AB
#> LA           0.6142827  1.962946     0.6534849      0.970378 -0.1573632
#> ELISA        0.5490645  2.316770     0.6534849      0.970378 -0.1573632
#> Nephelometry 1.2135916  1.598491     0.6534849      0.970378 -0.1573632
#>
#> $subgroups
#> [1] "LA"           "ELISA"        "Nephelometry"
```

How do I get a summary plot of the Rutter and Gatsonis subgroup model?

```
plot(ruttergatsonissub,
     size="se",
     col=c("red","black","green"),
     scale=0.015)
```

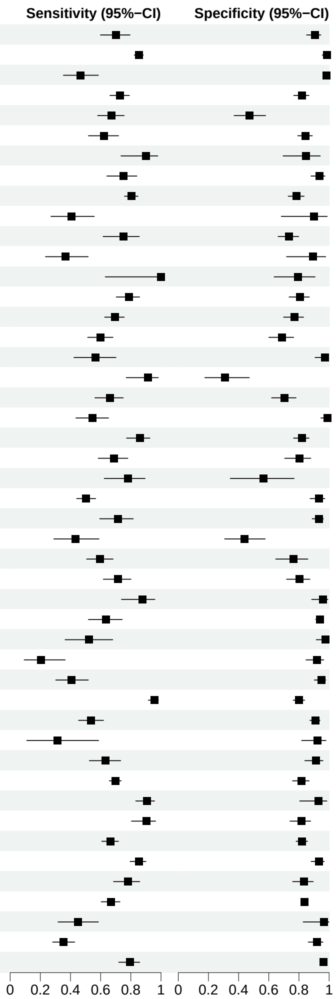


How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(ruttergatsonissub,subgroup_label = "Method")
```

Study	Method	TP	FP	FN	TN	Sensitivity (95%−CI)	Specificity (95%−CI)
Aho 1999	LA	64	16	27	153	0.70 [0.60, 0.79]	0.91 [0.85, 0.94]
Anuradha 2005	LA	482	2	82	153	0.85 [0.82, 0.88]	0.99 [0.95, 1.00]
Banchuin 1992	ELISA	36	6	41	313	0.47 [0.35, 0.58]	0.98 [0.96, 0.99]
Bas 2003	ELISA	143	43	53	196	0.73 [0.66, 0.79]	0.82 [0.77, 0.87]
Berthelot 1995	LA	80	50	39	45	0.67 [0.58, 0.76]	0.47 [0.37, 0.58]
Bizzaro 2001	Nephelometry	61	36	37	196	0.62 [0.52, 0.72]	0.84 [0.79, 0.89]
Bombardieri 2004	Nephelometry	27	6	3	33	0.90 [0.73, 0.98]	0.85 [0.69, 0.94]
Carpenter 1989	ELISA	60	8	20	119	0.75 [0.64, 0.84]	0.94 [0.88, 0.97]
Choi 2005	LA	261	54	63	197	0.81 [0.76, 0.85]	0.78 [0.73, 0.83]
Cordonnier 1996	LA	20	2	29	18	0.41 [0.27, 0.56]	0.90 [0.68, 0.99]
Das 2004	Nephelometry	42	46	14	127	0.75 [0.62, 0.86]	0.73 [0.66, 0.80]
Davis 1989	ELISA	18	3	31	25	0.37 [0.23, 0.52]	0.89 [0.72, 0.98]
de Bois 1996	ELISA	8	8	0	31	1.00 [0.63, 1.00]	0.79 [0.64, 0.91]
De Rycke 2004	LA	93	28	25	118	0.79 [0.70, 0.86]	0.81 [0.73, 0.87]
Despres 1994	LA	143	39	63	130	0.69 [0.63, 0.76]	0.77 [0.70, 0.83]
Dubucquoi 2004	ELISA	84	41	56	90	0.60 [0.51, 0.68]	0.69 [0.60, 0.77]
Fernandez−Suarez 2005	Nephelometry	30	2	23	73	0.57 [0.42, 0.70]	0.97 [0.91, 1.00]
Girelli 2004	Nephelometry	32	29	3	13	0.91 [0.77, 0.98]	0.31 [0.18, 0.47]
Goldbach−Mansky 2000	Nephelometry	70	39	36	93	0.66 [0.56, 0.75]	0.70 [0.62, 0.78]
Gomes−Daudrix 1994	ELISA	48	1	40	99	0.55 [0.44, 0.65]	0.99 [0.95, 1.00]
Greiner 2005	Nephelometry	75	42	12	191	0.86 [0.77, 0.93]	0.82 [0.76, 0.87]
Grootenboer−Mignot 2004	Nephelometry	64	18	29	73	0.69 [0.58, 0.78]	0.80 [0.71, 0.88]
Hitchon 2004	Nephelometry	32	10	9	13	0.78 [0.62, 0.89]	0.57 [0.34, 0.77]
Jansen 2003	Nephelometry	130	8	128	113	0.50 [0.44, 0.57]	0.93 [0.87, 0.97]
Jonsson 1998	ELISA	50	14	20	191	0.71 [0.59, 0.82]	0.93 [0.89, 0.96]
Kamali 2005	LA	20	32	26	25	0.43 [0.29, 0.59]	0.44 [0.31, 0.58]
Kwok 2005	Nephelometry	77	16	52	52	0.60 [0.51, 0.68]	0.76 [0.65, 0.86]
Lee 2003	LA	73	22	29	90	0.72 [0.62, 0.80]	0.80 [0.72, 0.87]
Lopez−Hoyos 2004	Nephelometry	36	3	5	70	0.88 [0.74, 0.96]	0.96 [0.88, 0.99]
Rantapaa−Dahlqvist 2003	ELISA	49	23	28	359	0.64 [0.52, 0.74]	0.94 [0.91, 0.96]
Raza 2005	LA	22	2	20	80	0.52 [0.36, 0.68]	0.98 [0.91, 1.00]
Saraux 1995	LA	8	8	31	91	0.21 [0.09, 0.36]	0.92 [0.85, 0.96]
Saraux 2003	ELISA	35	8	51	149	0.41 [0.30, 0.52]	0.95 [0.90, 0.98]
Sauerland 2005	Nephelometry	161	89	7	360	0.96 [0.92, 0.98]	0.80 [0.76, 0.84]
Schellekens 2000	ELISA	80	28	69	284	0.54 [0.45, 0.62]	0.91 [0.87, 0.94]
Soderlin 2004	LA	5	4	11	49	0.31 [0.11, 0.59]	0.92 [0.82, 0.98]
Spiritus 2004	Nephelometry	57	9	33	93	0.63 [0.53, 0.73]	0.91 [0.84, 0.96]
Suzuki 2003	Nephelometry	383	38	166	170	0.70 [0.66, 0.74]	0.82 [0.76, 0.87]
Swedler	Nephelometry	89	3	9	39	0.91 [0.83, 0.96]	0.93 [0.81, 0.99]
Thammanichanond 2005	LA	57	25	6	111	0.90 [0.80, 0.96]	0.82 [0.74, 0.88]
Vallbracht 2004	ELISA	196	75	99	345	0.66 [0.61, 0.72]	0.82 [0.78, 0.86]
van Leeuwen 1988	ELISA	163	10	28	140	0.85 [0.80, 0.90]	0.93 [0.88, 0.97]
Vasiliauskiene 2001	ELISA	75	21	21	106	0.78 [0.69, 0.86]	0.83 [0.76, 0.89]
Visser 1996	ELISA	157	287	78	1466	0.67 [0.60, 0.73]	0.84 [0.82, 0.85]
Vittecoq 2001	LA	26	1	32	29	0.45 [0.32, 0.58]	0.97 [0.83, 1.00]
Vittecoq 2004	ELISA	62	11	114	127	0.35 [0.28, 0.43]	0.92 [0.86, 0.96]
Winkles 1989	LA	113	19	29	481	0.80 [0.72, 0.86]	0.96 [0.94, 0.98]



Notes For study-level covariates the design matrix Z need two identical consecutive rows per study, one for the diseased and one for the non-diseased.

```
RF2
#>                         study year  TP  FP  FN   TN  cutoff       method
#> 1                   Aho 1999 1999  64  16  27  153      NA           LA
#> 2              Anuradha 2005 2005 482   2  82  153   8.000           LA
#> 3              Banchuin 1992 1992  36   6  41  313      NA        ELISA
#> 4                   Bas 2003 2003 143  43  53  196      NA        ELISA
```

```
#> 5           Berthelot 1995 1995  80  50  39   45 100.000           LA
#> 6             Bizzaro 2001 2001  61  36  37  196      NA Nephelometry
#> 7         Bombardieri 2004 2004  27   6   3   33  15.000 Nephelometry
#> 8           Carpenter 1989 1989  60   8  20  119  87.000        ELISA
#> 9                Choi 2005 2005 261  54  63  197   9.000           LA
#> 10         Cordonnier 1996 1996  20   2  29   18  40.000           LA
#> 11                Das 2004 2004  42  46  14  127  16.300 Nephelometry
#> 12              Davis 1989 1989  18   3  31   25      NA        ELISA
#> 13            de Bois 1996 1996   8   8   0   31   3.000        ELISA
#> 14          De Rycke 2004 2004  93  28  25  118   3.125           LA
#> 15            Despres 1994 1994 143  39  63  130 100.000           LA
#> 16          Dubucquoi 2004 2004  84  41  56   90  20.000        ELISA
#> 17   Fernandez-Suarez 2005 2005  30   2  23   73  50.000 Nephelometry
#> 18            Girelli 2004 2004  32  29   3   13  20.000 Nephelometry
#> 19    Goldbach-Mansky 2000 2000  70  39  36   93  20.000 Nephelometry
#> 20      Gomes-Daudrix 1994 1994  48   1  40   99      NA        ELISA
#> 21            Greiner 2005 2005  75  42  12  191      NA Nephelometry
#> 22 Grootenboer-Mignot 2004 2004  64  18  29   73  20.000 Nephelometry
#> 23            Hitchon 2004 2004  32  10   9   13  20.000 Nephelometry
#> 24             Jansen 2003 2003 130   8 128  113  30.000 Nephelometry
#> 25            Jonsson 1998 1998  50  14  20  191  40.000        ELISA
#> 26             Kamali 2005 2005  20  32  26   25  20.000           LA
#> 27               Kwok 2005 2005  77  16  52   52  15.000 Nephelometry
#> 28                Lee 2003 2003  73  22  29   90  80.000           LA
#> 29        Lopez-Hoyos 2004 2004  36   3   5   70  22.000 Nephelometry
#> 32 Rantapaa-Dahlqvist 2003 2003  49  23  28  359  20.000        ELISA
#> 33               Raza 2005 2005  22   2  20   80  30.000           LA
#> 34             Saraux 1995 1995   8   8  31   91  40.000           LA
#> 35             Saraux 2003 2003  35   8  51  149      NA        ELISA
#> 36          Sauerland 2005 2005 161  89   7  360  20.000 Nephelometry
#> 37        Schellekens 2000 2000  80  28  69  284   3.000        ELISA
#> 38          Soderlin 2004 2004   5   4  11   49  20.000           LA
#> 39           Spiritus 2004 2004  57   9  33   93  20.000 Nephelometry
#> 40             Suzuki 2003 2003 383  38 166  170  15.000 Nephelometry
#> 41                Swedler   NA  89   3   9   39  20.000 Nephelometry
#> 42    Thammanichanond 2005 2005  57  25   6  111  20.000           LA
#> 43         Vallbracht 2004 2004 196  75  99  345  15.000        ELISA
#> 44        van Leeuwen 1988 1988 163  10  28  140      NA        ELISA
#> 45     Vasiliauskiene 2001 2001  75  21  21  106  17.000        ELISA
#> 46             Visser 1996 1996 157 287  78 1466   9.000        ELISA
#> 47          Vittecoq 2001 2001  26   1  32   29  80.000           LA
#> 48          Vittecoq 2004 2004  62  11 114  127  16.000        ELISA
```

```
#> 49           Winkles 1989 1989 113  19  29  481      NA           LA
Z  <- model.matrix(~method,data=RF2)
# For study level covariates we need to two identical consective
# rows per study (diseased and non-diseased).
Z2 <- Z[rep(seq_len(nrow(Z)), each = 2), , drop = FALSE]
Z_pred <- matrix(c(1,0,0,1,1,0,1,0,1),ncol=3,nrow=3,byrow=T)

ruttergatsonisreg <- fitRutterGatsonisReg(data=RF2,
                                          TP=TP,
                                          FP=FP,
                                          FN=FN,
                                          TN=TN,
                                          study=study,
                                          Z=Z2,
                                          Z_pred=Z_pred)
ruttergatsonisreg
#> $data
#>                     study  TP  TN  FP  FN  n1  n0      sens      spec
#> 1                Aho 1999  64 153  16  27  91 169 0.7032967 0.9053254
#> 2           Anuradha 2005 482 153   2  82 564 155 0.8546099 0.9870968
#> 3           Banchuin 1992  36 313   6  41  77 319 0.4675325 0.9811912
#> 4                Bas 2003 143 196  43  53 196 239 0.7295918 0.8200837
#> 5          Berthelot 1995  80  45  50  39 119  95 0.6722689 0.4736842
#> 6            Bizzaro 2001  61 196  36  37  98 232 0.6224490 0.8448276
#> 7        Bombardieri 2004  27  33   6   3  30  39 0.9000000 0.8461538
#> 8          Carpenter 1989  60 119   8  20  80 127 0.7500000 0.9370079
#> 9               Choi 2005 261 197  54  63 324 251 0.8055556 0.7848606
#> 10        Cordonnier 1996  20  18   2  29  49  20 0.4081633 0.9000000
#> 11               Das 2004  42 127  46  14  56 173 0.7500000 0.7341040
#> 12             Davis 1989  18  25   3  31  49  28 0.3673469 0.8928571
#> 13           de Bois 1996   8  31   8   0   8  39 1.0000000 0.7948718
#> 14          De Rycke 2004  93 118  28  25 118 146 0.7881356 0.8082192
#> 15           Despres 1994 143 130  39  63 206 169 0.6941748 0.7692308
#> 16         Dubucquoi 2004  84  90  41  56 140 131 0.6000000 0.6870229
#> 17  Fernandez-Suarez 2005  30  73   2  23  53  75 0.5660377 0.9733333
#> 18          Girelli 2004  32  13  29   3  35  42 0.9142857 0.3095238
#> 19   Goldbach-Mansky 2000  70  93  39  36 106 132 0.6603774 0.7045455
#> 20      Gomes-Daudrix 1994  48  99   1  40  88 100 0.5454545 0.9900000
#> 21           Greiner 2005  75 191  42  12  87 233 0.8620690 0.8197425
#> 22 Grootenboer-Mignot 2004  64  73  18  29  93  91 0.6881720 0.8021978
#> 23           Hitchon 2004  32  13  10   9  41  23 0.7804878 0.5652174
#> 24            Jansen 2003 130 113   8 128 258 121 0.5038760 0.9338843
```

```
#> 25           Jonsson 1998  50  191  14  20  70  205 0.7142857 0.9317073
#> 26            Kamali 2005  20   25  32  26  46   57 0.4347826 0.4385965
#> 27              Kwok 2005  77   52  16  52 129   68 0.5968992 0.7647059
#> 28               Lee 2003  73   90  22  29 102  112 0.7156863 0.8035714
#> 29       Lopez-Hoyos 2004  36   70   3   5  41   73 0.8780488 0.9589041
#> 30 Rantapaa-Dahlqvist 2003  49  359  23  28  77  382 0.6363636 0.9397906
#> 31              Raza 2005  22   80   2  20  42   82 0.5238095 0.9756098
#> 32            Saraux 1995   8   91   8  31  39   99 0.2051282 0.9191919
#> 33            Saraux 2003  35  149   8  51  86  157 0.4069767 0.9490446
#> 34         Sauerland 2005 161  360  89   7 168  449 0.9583333 0.8017817
#> 35       Schellekens 2000  80  284  28  69 149  312 0.5369128 0.9102564
#> 36          Soderlin 2004   5   49   4  11  16   53 0.3125000 0.9245283
#> 37          Spiritus 2004  57   93   9  33  90  102 0.6333333 0.9117647
#> 38            Suzuki 2003 383  170  38 166 549  208 0.6976321 0.8173077
#> 39              Swedler  89   39   3   9  98   42 0.9081633 0.9285714
#> 40   Thammanichanond 2005  57  111  25   6  63  136 0.9047619 0.8161765
#> 41        Vallbracht 2004 196  345  75  99 295  420 0.6644068 0.8214286
#> 42       van Leeuwen 1988 163  140  10  28 191  150 0.8534031 0.9333333
#> 43    Vasiliauskiene 2001  75  106  21  21  96  127 0.7812500 0.8346457
#> 44            Visser 1996 157 1466 287  78 235 1753 0.6680851 0.8362807
#> 45          Vittecoq 2001  26   29   1  32  58   30 0.4482759 0.9666667
#> 46          Vittecoq 2004  62  127  11 114 176  138 0.3522727 0.9202899
#> 47           Winkles 1989 113  481  19  29 142  500 0.7957746 0.9620000
#>    recordid
#> 1         1
#> 2         2
#> 3         3
#> 4         4
#> 5         5
#> 6         6
#> 7         7
#> 8         8
#> 9         9
#> 10       10
#> 11       11
#> 12       12
#> 13       13
#> 14       14
#> 15       15
#> 16       16
#> 17       17
#> 18       18
```

```
#> 19       19
#> 20       20
#> 21       21
#> 22       22
#> 23       23
#> 24       24
#> 25       25
#> 26       26
#> 27       27
#> 28       28
#> 29       29
#> 30       30
#> 31       31
#> 32       32
#> 33       33
#> 34       34
#> 35       35
#> 36       36
#> 37       37
#> 38       38
#> 39       39
#> 40       40
#> 41       41
#> 42       42
#> 43       43
#> 44       44
#> 45       45
#> 46       46
#> 47       47
#>
#> $fit
#> $fit$par
#>   accuracy_coef   accuracy_coef   accuracy_coef  threshold_coef  threshold_coef
#>       2.4563116       0.2485221       0.3314158      -0.5500508      -0.1962501
#>  threshold_coef      shape_coef log_sigma_alpha log_sigma_theta
#>       0.4958570       0.1976819       0.1226137      -0.3702787
#>
#> $fit$objective
#> [1] 376.8612
#>
#> $fit$convergence
#> [1] 0
```

```
#>
#> $fit$iterations
#> [1] 27
#>
#> $fit$evaluations
#> function gradient
#>       39       28
#>
#> $fit$message
#> [1] "relative convergence (4)"
#>
#>
#> $sdreport
#> sdreport(.) result
#>                  Estimate Std. Error
#> accuracy_coef    2.4563116  0.3235767
#> accuracy_coef    0.2485221  0.4395407
#> accuracy_coef    0.3314158  0.4426001
#> threshold_coef  -0.5500508  0.2133415
#> threshold_coef  -0.1962501  0.2609639
#> threshold_coef   0.4958570  0.2622323
#> shape_coef       0.1976819  0.1698395
#> log_sigma_alpha  0.1226137  0.1207141
#> log_sigma_theta -0.3702787  0.1189556
#> Maximum gradient component: 0.0005885859
#>
#> $sdreport2
#>                  Estimate Std. Error
#> accuracy_coef   2.45631156 0.32357673
#> accuracy_coef   0.24852208 0.43954066
#> accuracy_coef   0.33141584 0.44260014
#> threshold_coef -0.55005084 0.21334149
#> threshold_coef -0.19625009 0.26096393
#> threshold_coef  0.49585695 0.26223232
#> shape_coef      0.19768191 0.16983950
#> shape_coef      0.00000000 0.00000000
#> shape_coef      0.00000000 0.00000000
#> sigma2_alpha    1.27791200 0.30852400
#> sigma2_theta    0.47684805 0.11344753
#> Lambda_Pred     2.45631156 0.32357673
#> Lambda_Pred     2.70483365 0.32695391
#> Lambda_Pred     2.78772741 0.30577879
```

```
#> Theta_Pred     -0.55005084 0.21334149
#> Theta_Pred     -0.74630093 0.20991415
#> Theta_Pred     -0.05419388 0.21213237
#> beta_Pred       0.19768191 0.16983950
#> beta_Pred       0.19768191 0.16983950
#> beta_Pred       0.19768191 0.16983950
#> logitsens       0.82617129 0.28629952
#> logitsens       1.05130416 0.28606887
#> logitsens       1.12639652 0.27689490
#> sens            0.69554476 0.06062743
#> sens            0.74102525 0.05489857
#> sens            0.75517328 0.05119416
#>
#> $sensspec
#>        spec conflevel logitsens Std_Error  CI_Lower CI_Upper      Sens
#> 1 0.8461538      0.95 0.8261713 0.2862995 0.2650345 1.387308 0.6955448
#> 2 0.8461538      0.95 1.0513042 0.2860689 0.4906195 1.611989 0.7410253
#> 3 0.8461538      0.95 1.1263965 0.2768949 0.5836925 1.669101 0.7551733
#>   SensCI_Lower SensCI_Upper
#> 1    0.5658735    0.8001621
#> 2    0.6202524    0.8336873
#> 3    0.6419166    0.8414559
#>
#> attr(,"class")
#> [1] "RutterGatsonisReg"
```

References

Reitsma, J. B., et al. (2005). Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *Journal of Clinical Epidemiology*, 58(10), 982–990.

Rutter, C. M., & Gatsonis, C. A. (2001). A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Statistics in Medicine*, 20(19), 2865–2884.

Harbord, R. M., Deeks, J. J., Egger, M., Whiting, P., & Sterne, J. A. C. (2007). A unification of models for meta-analysis of diagnostic accuracy studies. *Biostatistics*, 8(2), 239–251.

Riley, R. D., Ensor, J., Jackson, D., & Burke, D. L. (2018). Deriving percentage study weights in multi-parameter meta-analysis models. *Statistical Methods in Medical Research*, 27(10), 2885–2905.

Hoyer, A., Hirt, S., Kuss, O. (2018). Meta-analysis of full ROC curves using bivariate time-to-event models for interval-censored data. *Research Synthesis Methods*, 9(1), 62-72.

Deeks, J. J., Bossuyt, P. M., Leeflang, M. M., & Takwoingi, Y. (editors) (2023). Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy. Version 2.0 (updated July 2023). Cochrane.